Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/24/2006
PI Name:	Maris, Humphrey J. Ph.D.		
Project Title:	Coalescence of Superfluid Helium Drops in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02912	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Seidel, George (Brown University)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We are conducting ground-based research to study the coalescence of levitated liquid drops. Drops of liquid helium are levitated using a superconducting magnet levitation system developed in our laboratory. Drops of classical fluids are levitated by immersion in a second fluid of nearly the same density. We have made a detailed investigation of the initial rate of growth of the neck between the two drops. The results of this research will provide necessary data for the design of an experiment to study coalescence of large superfluid helium drops in space. The long-term objective of this research is the understanding of the process of coalescence of liquid drops. The unique properties of helium make it an ideal material for coalescence studies. The hydrodynamics of helium in the superfluid phase is described by the two-fluid model of Landau and Tisza. At sufficiently low temperatures where the density of the normal fluid becomes extremely small, liquid helium acts as an ideal fluid with no viscosity. Our long term goal is a		

	space experiment to study the coalescence of superfluid helium in this low temperature regime. In a space experiment large helium drops could be used, and this would make it possible to resolve the early stages of the coalescence process to a degree that is impossible in a ground based experiment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study of coalescence is of great importance in a number of practical applications, including the sintering process, the combustion of fuel sprays, and the process of spray painting. In nature, coalescence affects the distribution of size of rain drops.
Task Progress:	We are using the helium levitation cryostat from our previous work to study the breakdown of a droplet of superfluid helium with a large positive charge. As positive charge is added to a droplet, it distributes itself around the surface to minimize the Coulomb repulsion. When the effect of Coulomb repulsion exceeds the surface tension, the droplet has reached a critical charge and a breakup will result. The critical charge is given by Q=8 \pi\sqrt(\sigma r^3) where \sigma is the surface tension and r is the radius of the drop. A previous experiment (D. Duft et al. Physical Review Letters 89, 084503 2002, and Nature 421, 128 2003) has seen droplets of ethylene glycol break apart by way of Rayleigh jets. As the droplet neared its critical charge, two small cones formed at either end of the droplet. From these cones, a large number of very small droplets was ejected. These droplets carried a large percentage of the charge (~30%) but only a small percentage of the volume (~2%). In our present work we aim to visually record the breakdown of a droplet of superfluid helium with a high speed camera and study the relation of critical charge to drop radius. We want to explore what differences, if any, there are between the breakup of a superfluid drop compared to a normal viscous drop. We have explored two methods of charging a droplet of helium. The first is through the placement of an alpha source within the droplet chamber. We have two concerns with this approach. The first is that the alpha particles could ionize gaseous helium within the chamber. The droplet, once it has picked up a positive charge, attracts the free electrons and repels the He+ ions. This reduces the charge on the droplet and could lead to a maximum charge. We can avoid this by cooling the chamber to 500 mK or below where the vapor pressure is small enough that the likelihood of ionization is small. More problematic, however, is the creation of excited states with the droplet itself. As the alpha particle deposits its energy into the helium droplet, it creates a large amount ionizations. Research in our lab has shown that some of these electron/ion pairs recombine to form relatively stable dimers, pairs of atoms which share an energetic electron. These dimers migrate to the surface where they are loosely bound. At the surface, their decay can eject either (or both) the electron or the ion into the vacuum. Once the droplet acquires a positive charge, the electrons will be preferentially reabsorbed while the positive ions will be ejected. This will lead to a maximum acquired charge. The second method is to create ions directly out of an electrode covered by a helium film. Previous attempts have managed to charge the droplet to 10^5 to 10^6 charge carriers. The maximum charge was limited because the relatively large, and nearby, electrodes exerted large forces on a charged drop. Any attempt to further charge the drop pushed the drop out of its equilibrium position of the magnetic trap. Our current setup uses a smaller tip to ionize and drive the helium film which should not unduly disturb the droplet.
Bibliography Type:	Description: (Last Updated: 01/07/2020)